Physical Properties of Rocks and How We Estimate Them

Derrick Hasterok

May 27, 2026

1 Introduction

In geophysics we typically *measure fields* and *infer properties*, the reverse of many classical physics problems. Rock properties arise from minerals and the fluids in pore space and depend on the *state* of the system (pressure, temperature, volume, composition).

1.1 State variables

- Pressure P (Pa, MPa, GPa, kbar). Note: $1\,\mathrm{kbar} \approx 100\,\mathrm{MPa}$.
- Temperature T (K or °C). Absolute scale: not "degrees K"; conversion: $T_\mathrm{K} = T_{^\circ\mathrm{C}} + 273.15$.
- Volume V (m^3 or km^3).
- Composition (moles, wt%, ppm, fractions). Can refer to phase modes or elemental/oxide concentrations.

2 Key Physical Properties

- Density: $\rho = m/V$ ($\mathrm{kg\,m^{-3}}$).
- Thermal expansivity (volumetric, at P const.): α ($\mathrm{K^{-1}}$).
- Compressibility β_T and bulk moduli $K_T = 1/\beta_T$, K_S (MPa, GPa).
- Shear modulus μ (MPa, GPa).
- Seismic velocities: v_p (compressional), v_s (shear) ($\mathrm{m\,s^{-1}}$ or $\mathrm{km\,s^{-1}}$).
- Thermal conductivity k ($\mathrm{W\,m^{-1}\,K^{-1}}$); heat capacity c_p ($\mathrm{J\,kg^{-1}\,K^{-1}}$).
- Thermal diffusivity $\kappa = k/(\rho c_p)$ ($\mathrm{mm^2\,s^{-1}}$ or $\mathrm{km^2\,Ma^{-1}}$).
- Heat production A ($\mathrm{W\,m^{-3}}$ or $\mu\mathrm{W\,m^{-3}}$), or H ($\mathrm{W\,kg^{-1}}$).

3 Strategies to Estimate Properties

3.1 Direct measurements

- Lab or in situ; strong control on state; targeted protocols.
- Consider scaling, representativeness, cost, and sampling.

3.2 From composition

- From mineral modes: simple to apply when modes known; extrapolation outside calibrations is risky (??).
- From major elements: widely available; captures broad trends but may obscure mineral-control (assumptions needed).
- Thermodynamics: internally consistent; works with or without known modes; computationally intensive and specialized.

3.3 Indirect correlations

- Property–property regressions (e.g., v_p vs ρ); rapid domain mapping but potentially non-unique.
- Data-driven/ML: handles large, multivariate datasets; may be biased by training data and limited in state extrapolation.

4 Examples and Figures from Notes

4.1 Density from composition

An empirical relation for many silicates (Group I) is suggested as

$$\rho\,[\mathrm{kg\,m^{-3}}] = 2532 + 216\,\mathrm{Fe} + 608\,M - 10\,\mathrm{MALI}, \quad \text{(notation as in notes; confirm definitions: ??)} \tag{1}$$

where Fe is an iron number (wt%), M a maficity parameter, and MALI is the modified alkali–lime index (precise definitions deferred: ??).

4.2 Other property trends

Several properties co-vary with composition or with each other (e.g., thermal conductivity with composition; weakly with v_p), enabling indirect estimates.

5 Pressure, Temperature, and Porosity Effects

For a given rock, three principal factors influence velocity:

- Pressure increases close cracks and stiffens the frame $\Rightarrow$ v increases.
- Temperature increases reduce density and moduli $\Rightarrow$ v decreases.
- Porosity/fractures impede wave paths $\Rightarrow$ lower apparent v (in crystalline rocks, density reduction may be minor).

6 Mixing Relations

Rocks are mixtures; bulk properties require mixing laws.

6.1 Common means

- Arithmetic mean (e.g., density, some heat capacities): $X = \sum_i w_i X_i$.
- Geometric mean (random mixtures; often used for conductivity): $X = \prod_i X_i^{w_i}$.
- Harmonic mean (series layers; transport-limited): $X = \left(\sum_i \frac{w_i}{X_i}\right)^{-1}$.

6.2 Voigt–Reuss–Hill (VRH)

For elastic moduli/velocities: X_V (arithmetic), X_R (harmonic), and Hill average $X_{\mathrm{VRH}} = \frac{1}{2}(X_V + X_R)$.

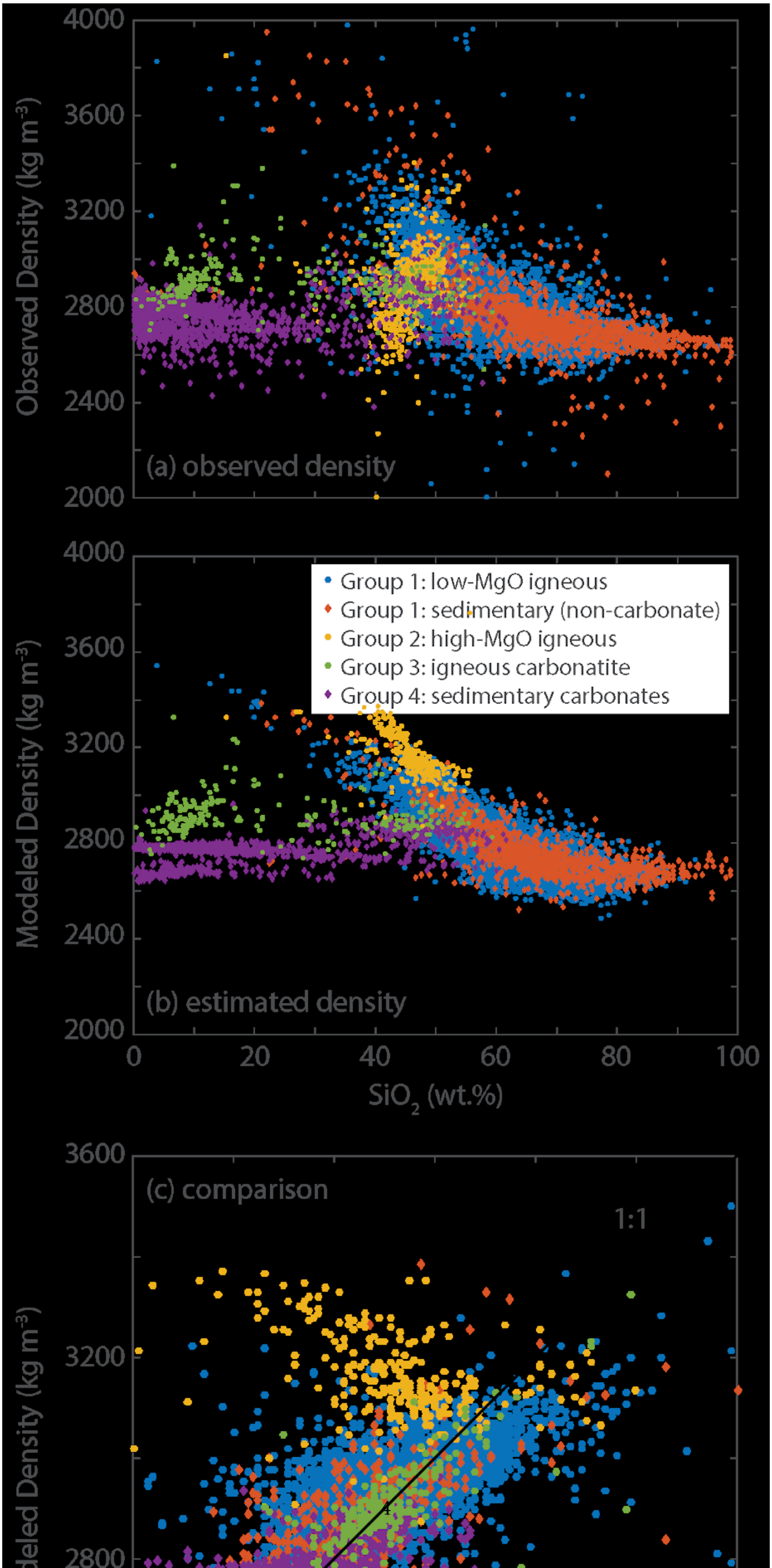
6.3 Isotropy vs anisotropy

- Isotropic properties (e.g., density) are direction independent.
- Anisotropic properties (e.g., thermal/electrical conductivity, elastic moduli) vary with direction.

Bulk density as an example. With volume fractions f_i and constituent densities ρ_i,

$$\rho_{\text{bulk}} = \sum_i f_i \, \rho_i. \tag{2}$$

oindent*Transcription notes:* Ambiguities in the handwriting/OCR are marked with “??” to flag items needing your confirmation.



Observed Density (kg m⁻³)
4000
3600
3200
2800
2400
2000
(a) observed density
Modeled Density (kg m⁻³)
4000
3600
3200
2800
2400
2000
Group 1: low-MgO igneous
Group 1: sedimentary (non-carbonate)
Group 2: high-MgO igneous
Group 3: igneous carbonatite
Group 4: sedimentary carbonates
(b) estimated density
0
20
40
60
80
100
SiO2 (wt.%)
3600
3200
2800
(c) comparison
1:1

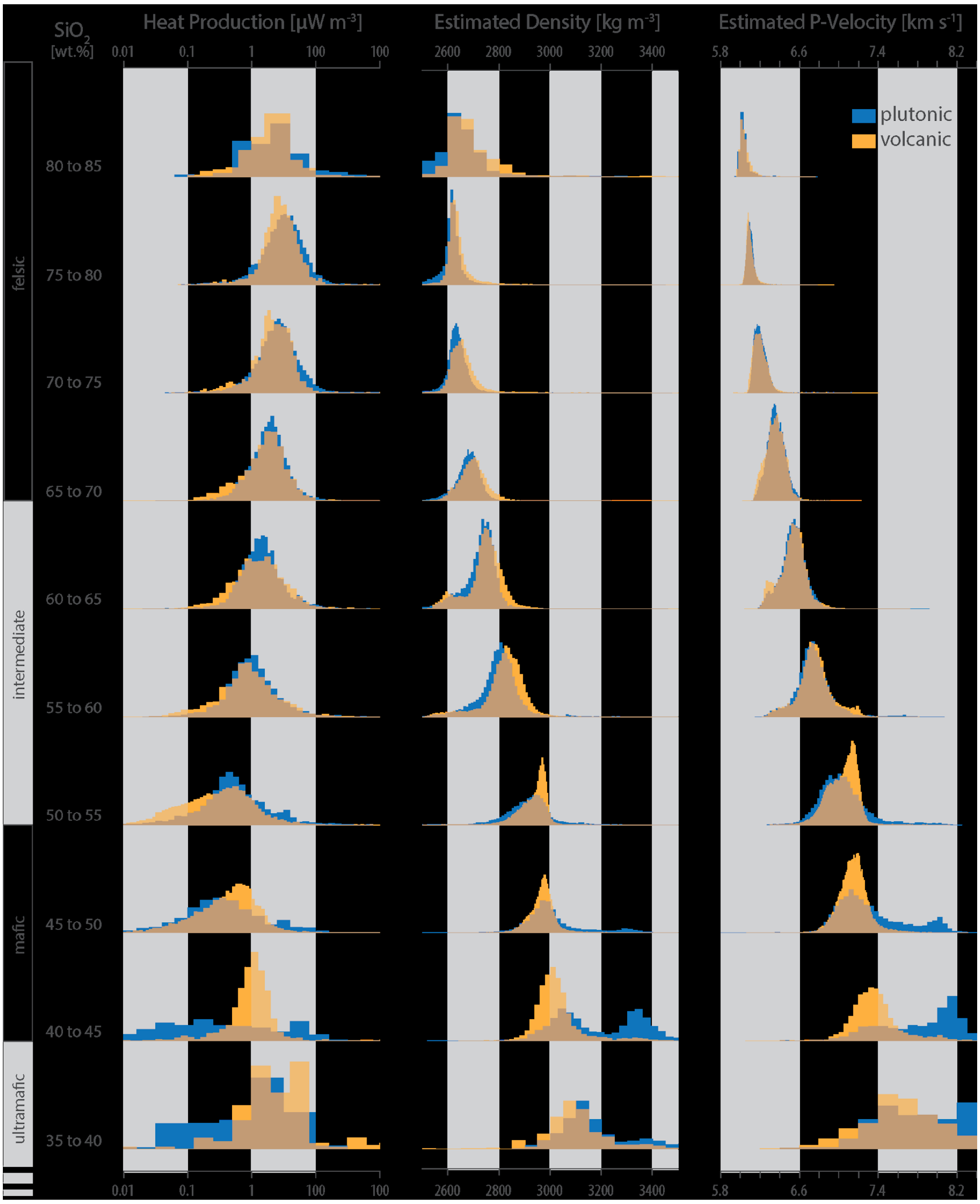


Figure 2: Compositional trends among properties (extracted from notes).

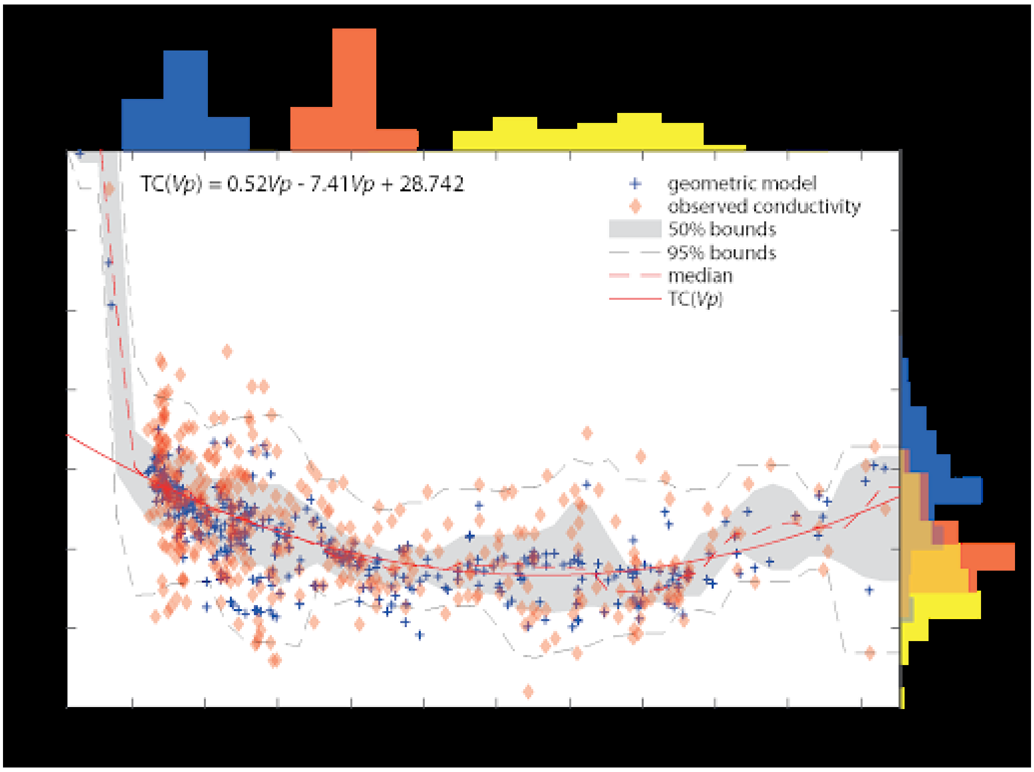


Figure 3: Thermal conductivity vs composition or v_p (extracted from notes).

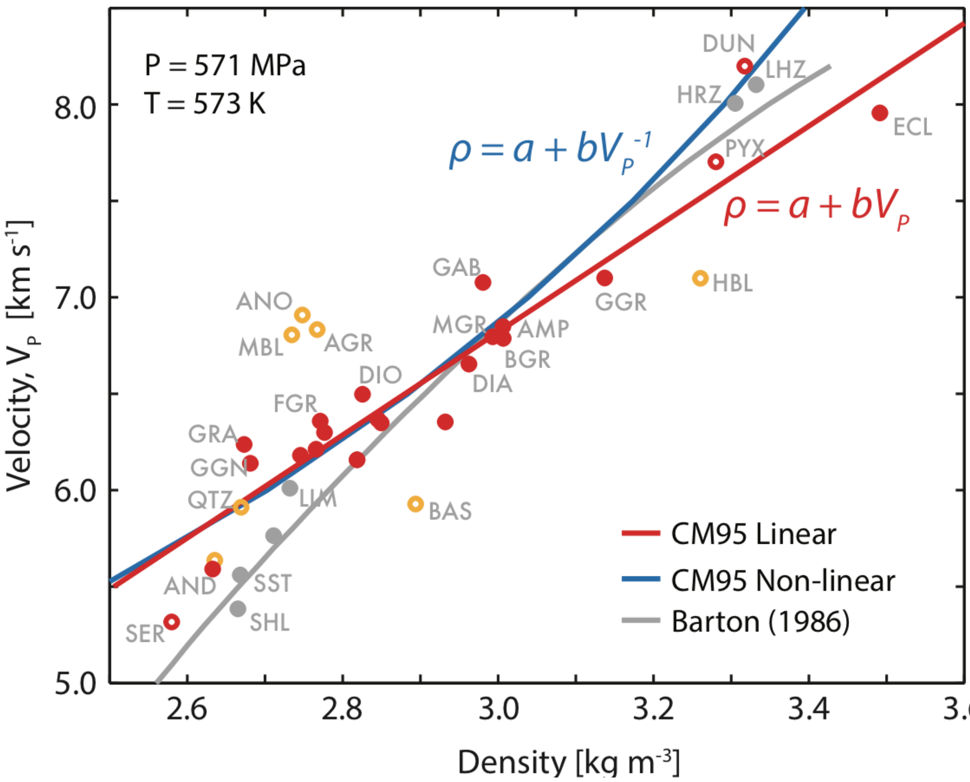


Figure 4: Seismic velocity vs density as composition proxy (extracted from notes).

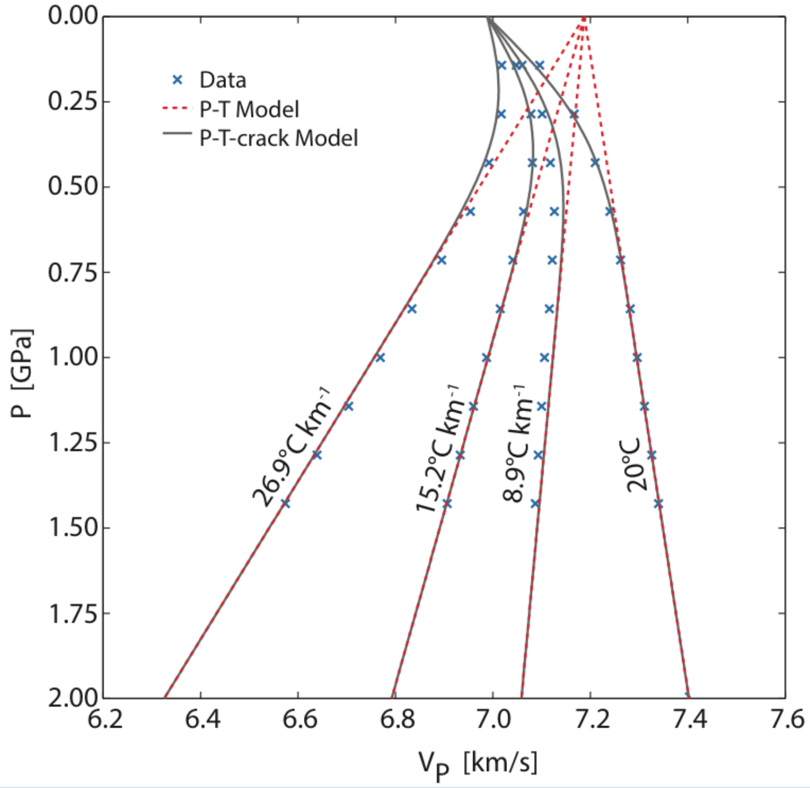


Figure 5: Velocity trends with P and T (extracted from notes).

Figure 6: Effect of porosity/fractures on velocity (extracted from notes).

Figure 7: Additional schematic/plot related to mixing or trends (extracted from notes).